Experiment No. 2

Exploratory Data Analysis (EDA) on a Real-World Business Dataset using Python

Aim

To perform Exploratory Data Analysis on the IBM HR Employee Attrition dataset using Python in order to understand its structure, identify trends, detect anomalies, analyse feature relationships and generate business insights through descriptive statistics and visualisations.

Requirements

- Python 3, Jupyter Notebook / Google Colab; Pandas, NumPy, Matplotlib, Seaborn
- Dataset: IBM HR Employee Attrition - 1,470 employees, 35 columns. Business objective: understand why employees leave (*Attrition* = Yes/No).

Procedure

- Select the dataset, import libraries and load the data.
- Explore dimensions, data types and summary statistics (`info()`, `describe()`, `value_counts()`).
- Plot histograms/density plots of numerical features; count plots of categorical features.
- Build the correlation matrix and heatmap; draw box plots and scatter plots.
- Perform univariate, bivariate and multivariate analysis.
- Summarise key findings and give recommendations.

Program and Output

The program was run on the actual dataset; green boxes show the real output and the figures are the real plots produced.

Steps 2-3: Load the data and explore its structure

```
import pandas as pd
import numpy as np
import matplotlib
matplotlib.use("Agg")                       # remove this line in Jupyter/Colab
import matplotlib.pyplot as plt
import seaborn as sns

pd.set_option("display.width", 120)
pd.set_option("display.max_columns", 20)
sns.set_style("whitegrid")

# ---------- Step 2-3: load data and explore structure ----------
df = pd.read_csv("hr.csv", encoding="utf-8-sig")      # IBM HR Employee Attrition
print("Shape:", df.shape)
print("Missing values:", df.isnull().sum().sum(), "| Duplicate rows:", df.duplicated().sum())
print("Constant columns:", list(df.columns[df.nunique() == 1]))
print("Numeric columns:", df.select_dtypes("number").shape[1],
      "| Categorical columns:", df.select_dtypes(exclude="number").shape[1])
print(df[["Age", "MonthlyIncome", "DistanceFromHome", "YearsAtCompany", "TotalWorkingYears"]].describe().round(2))
print("\nAttrition counts:\n", df["Attrition"].value_counts())
print("Attrition %:\n", (df["Attrition"].value_counts(normalize=True) * 100).round(1))

df = df.drop(columns=["EmployeeCount", "Over18", "StandardHours", "EmployeeNumber"])  # no information
df["Attrition_flag"] = (df["Attrition"] == "Yes").astype(int)
num_cols = df.select_dtypes("number").columns.drop("Attrition_flag")
```

Output:

```
Shape: (1470, 35)
Missing values: 0 | Duplicate rows: 0
Constant columns: ['EmployeeCount', 'Over18', 'StandardHours']
Numeric columns: 26 | Categorical columns: 9
            Age  MonthlyIncome  DistanceFromHome  YearsAtCompany  TotalWorkingYears
count  1470.00        1470.00           1470.00         1470.00            1470.00
```

```
mean      36.92        6502.93             9.19            7.01             11.28
std        9.14        4707.96             8.11            6.13              7.78
min       18.00        1009.00             1.00            0.00              0.00
25%       30.00        2911.00             2.00            3.00              6.00
50%       36.00        4919.00             7.00            5.00             10.00
75%       43.00        8379.00            14.00            9.00             15.00
max       60.00       19999.00            29.00           40.00             40.00

Attrition counts:
 Attribution
No     1233
Yes     237
Name: count, dtype: int64
Attrition %:
 Attribution
No     83.9
Yes    16.1
Name: proportion, dtype: float64
```

Step 4: Distribution of numerical features

```
# ---------- Step 4: distribution of numerical features ----------
fig, ax = plt.subplots(2, 2, figsize=(10, 7))
for a, c in zip(ax.ravel(), ["Age", "MonthlyIncome", "DistanceFromHome", "YearsAtCompany"]):
    sns.histplot(df[c], kde=True, bins=30, ax=a, color="steelblue")
    a.axvline(df[c].mean(), color="red", ls="--", label="mean")
    a.axvline(df[c].median(), color="green", ls="-", label="median")
    a.set_title(c); a.legend()
plt.tight_layout(); plt.savefig("fig1_hist.png", dpi=110); plt.show()
print("Skewness:\n", df[["Age", "MonthlyIncome", "DistanceFromHome", "YearsAtCompany"]].skew().round(2))
```

Output:

```
Skewness:
 Age                0.41
MonthlyIncome       1.37
DistanceFromHome    0.96
YearsAtCompany      1.76
dtype: float64
```

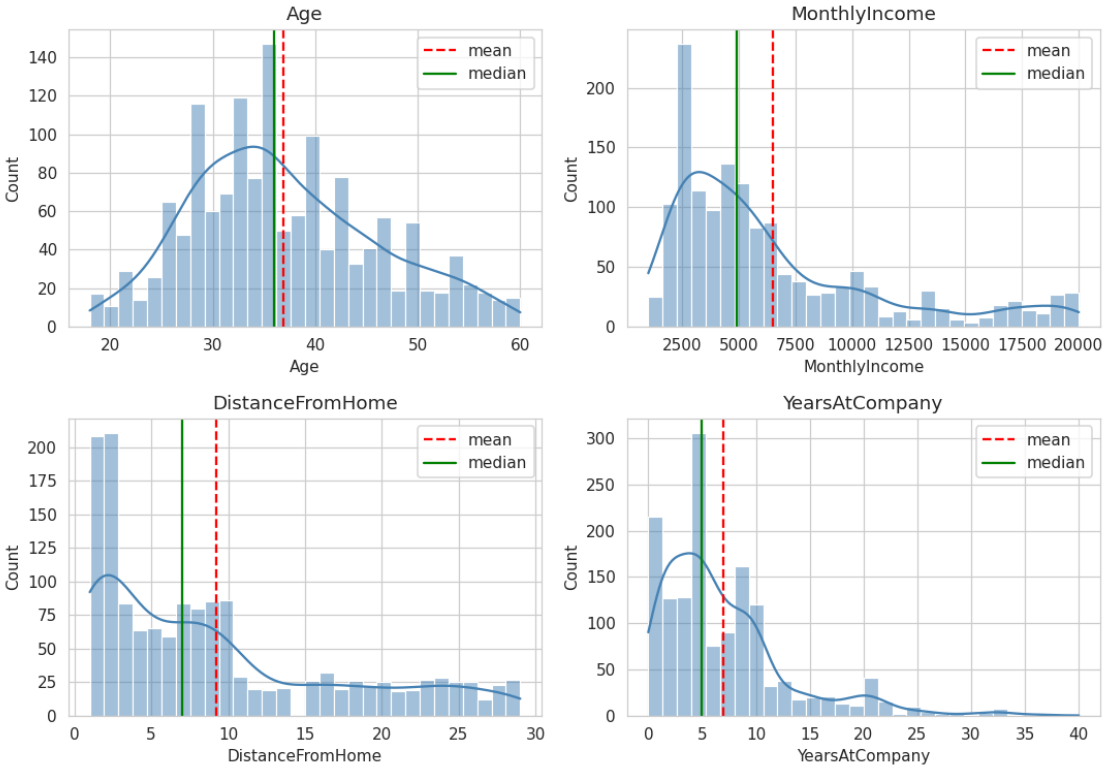


Fig. 1: Histograms with KDE (red dashed = mean, green = median)

Step 5: Categorical variables

```
# ---------- Step 5: categorical variables ----------
fig, ax = plt.subplots(2, 2, figsize=(11, 7))
sns.countplot(x="Attrition", data=df, ax=ax[0, 0], palette="Set2", hue="Attrition", legend=False)
sns.countplot(x="Department", data=df, ax=ax[0, 1], palette="Set2", hue="Department", legend=False)
```

```
sns.countplot(x="BusinessTravel", data=df, ax=ax[1, 0], palette="Set2", hue="BusinessTravel", legend=False)
sns.countplot(y="JobRole", data=df, ax=ax[1, 1], order=df["JobRole"].value_counts().index,
              palette="Set2", hue="JobRole", legend=False)
for a in (ax[0, 1], ax[1, 0]): a.tick_params(axis="x", rotation=15)
plt.tight_layout(); plt.savefig("fig2_count.png", dpi=110); plt.show()
print(df["Department"].value_counts(), "\n")
print(df["JobRole"].value_counts().head(3))
```

Output:

```
Department
Research & Development    961
Sales                     446
Human Resources            63
Name: count, dtype: int64

JobRole
Sales Executive          326
Research Scientist       292
Laboratory Technician    259
Name: count, dtype: int64
```

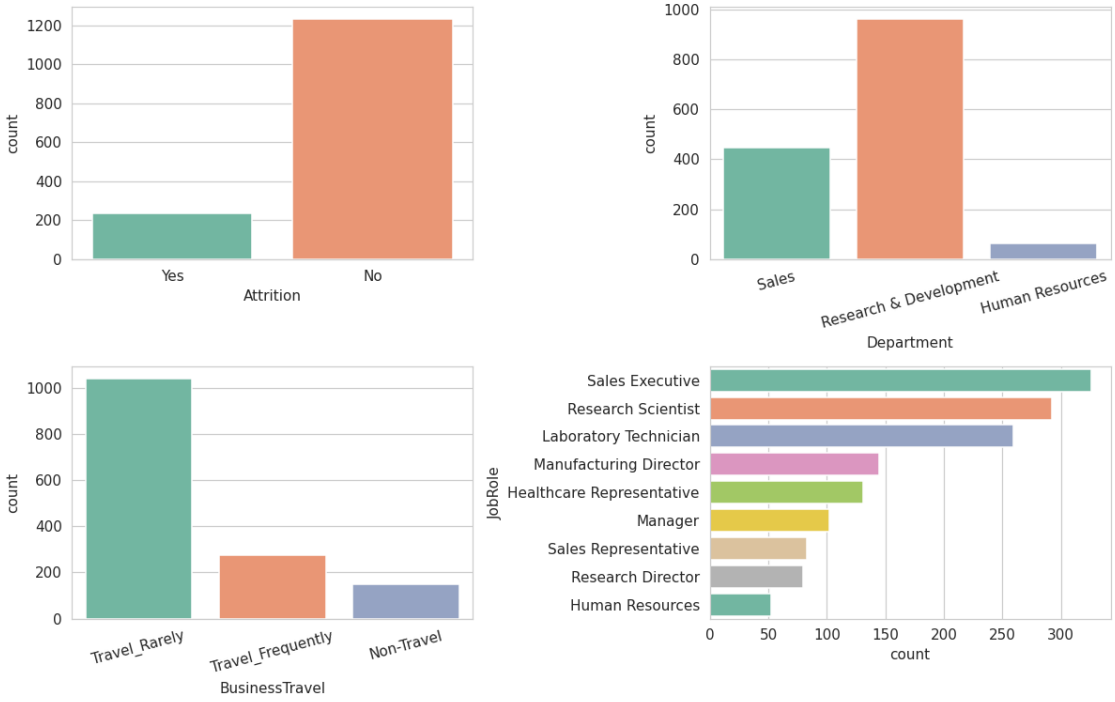


Fig. 2: Count plots of categorical variables

Step 6: Correlation matrix and heatmap

```
# ---------- Step 6: correlation matrix and heatmap ----------
cols = ["Age", "MonthlyIncome", "JobLevel", "TotalWorkingYears", "YearsAtCompany",
        "YearsInCurrentRole", "YearsWithCurrManager", "DistanceFromHome",
        "JobSatisfaction", "WorkLifeBalance", "Attrition_flag"]
corr = df[cols].corr()
plt.figure(figsize=(9, 7))
sns.heatmap(corr, annot=True, fmt=".2f", cmap="coolwarm", center=0, square=True)
plt.title("Correlation heatmap"); plt.tight_layout(); plt.savefig("fig3_heatmap.png", dpi=110); plt.show()

pairs = corr.where(np.triu(np.ones(corr.shape, dtype=bool), 1)).stack()
print("Highly correlated pairs (|r| > 0.7):")
print(pairs[pairs.abs() > 0.7].sort_values(ascending=False).round(2))
print("\nCorrelation with Attrition_flag:")
print(corr["Attrition_flag"].drop("Attrition_flag").sort_values().round(3))
```

Output:

```
Highly correlated pairs (|r| > 0.7):
MonthlyIncome       JobLevel               0.95
JobLevel            TotalWorkingYears      0.78
MonthlyIncome       TotalWorkingYears      0.77
YearsAtCompany      YearsWithCurrManager   0.77
                    YearsInCurrentRole     0.76
YearsInCurrentRole  YearsWithCurrManager   0.71
dtype: float64
```

```
Correlation with Attrition_flag:
TotalWorkingYears      -0.171
JobLevel               -0.169
YearsInCurrentRole     -0.161
MonthlyIncome          -0.160
Age                    -0.159
YearsWithCurrManager   -0.156
YearsAtCompany         -0.134
JobSatisfaction        -0.103
WorkLifeBalance        -0.064
DistanceFromHome        0.078
Name: Attrition_flag, dtype: float64
```

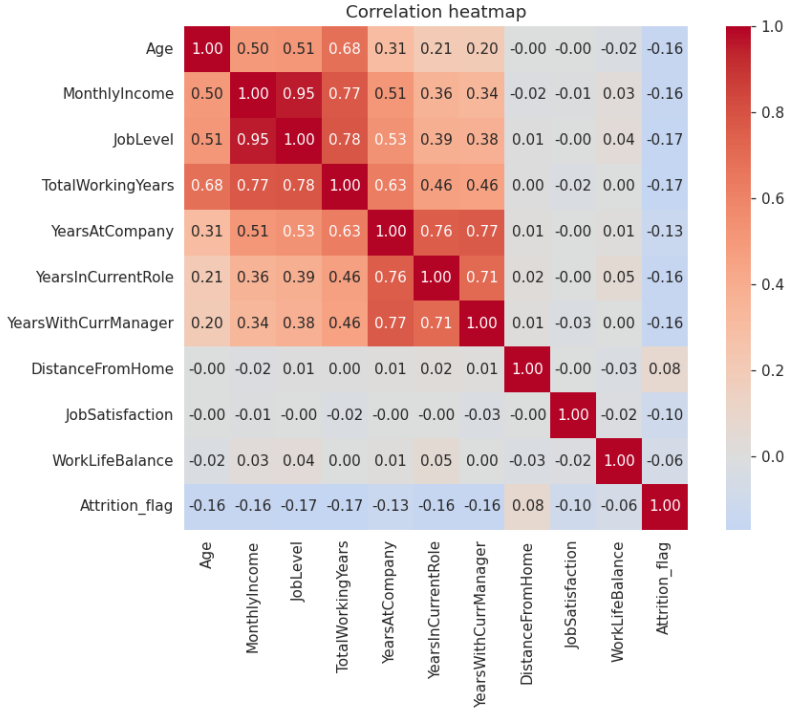


Fig. 3: Correlation heatmap

Step 7: Box plots and scatter plots

```
# ---------- Step 7: box plots and scatter plots ----------
fig, ax = plt.subplots(2, 2, figsize=(11, 8))
sns.boxplot(x="Attrition", y="MonthlyIncome", data=df, ax=ax[0, 0], palette="Set2", hue="Attrition", legend=False)
sns.boxplot(x="Attrition", y="Age", data=df, ax=ax[0, 1], palette="Set2", hue="Attrition", legend=False)
sns.scatterplot(x="TotalWorkingYears", y="MonthlyIncome", hue="Attrition", data=df, ax=ax[1, 0], alpha=0.6)
sns.scatterplot(x="Age", y="MonthlyIncome", hue="Attrition", data=df, ax=ax[1, 1], alpha=0.6)
plt.tight_layout(); plt.savefig("fig4_box_scatter.png", dpi=110); plt.show()

q1, q3 = df["MonthlyIncome"].quantile([0.25, 0.75]); iqr = q3 - q1
print("MonthlyIncome outliers (IQR):", int((df["MonthlyIncome"] > q3 + 1.5 * iqr).sum()))
print(df.groupby("Attrition")[["MonthlyIncome", "Age", "YearsAtCompany", "TotalWorkingYears"]].median())
```

Output:

```
MonthlyIncome outliers (IQR): 114
           MonthlyIncome   Age  YearsAtCompany  TotalWorkingYears
Attrition
No                5204.0  36.0             6.0               10.0
Yes               3202.0  32.0             3.0                7.0
```

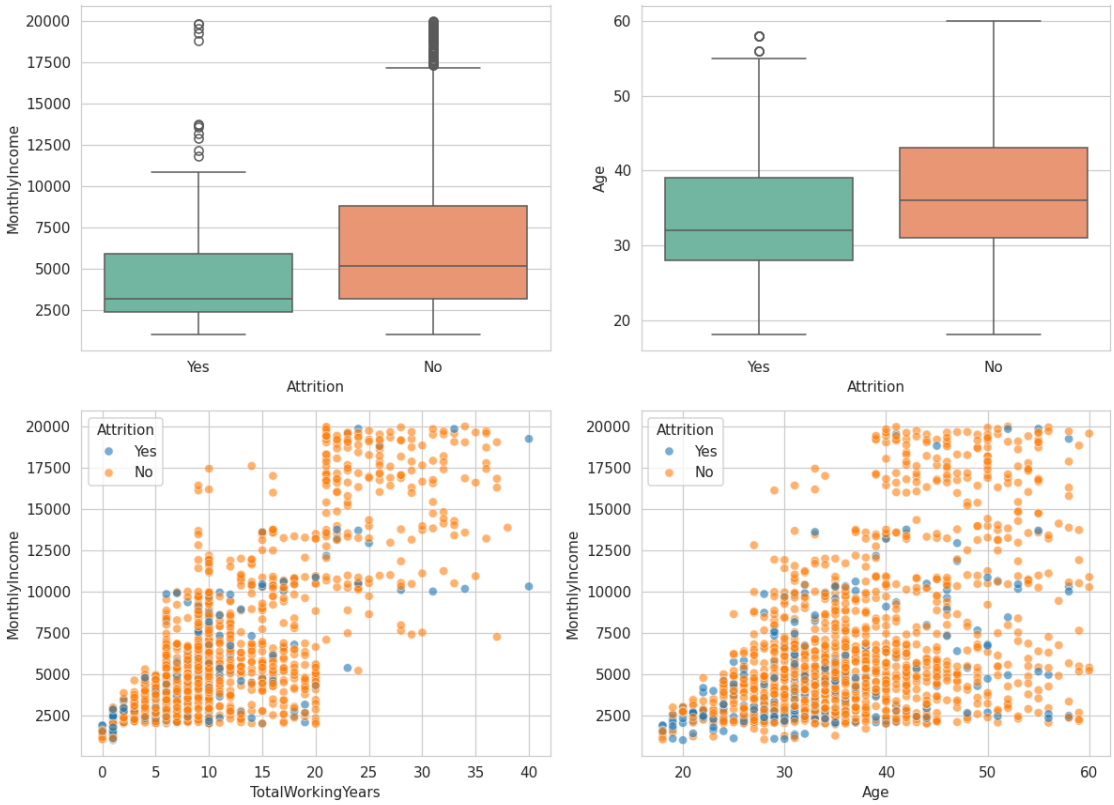


Fig. 4: Box plots and scatter plots by Attrition

Step 8: Univariate and bivariate analysis

```
# ---------- Step 8a: univariate + bivariate (attrition rate by category) ----------
rates = {}
fig, ax = plt.subplots(2, 2, figsize=(11, 8))
for a, c in zip(ax.ravel(), ["OverTime", "MaritalStatus", "BusinessTravel", "JobRole"]):
    r = (df.groupby(c)["Attrition_flag"].mean() * 100).sort_values(ascending=False)
    rates[c] = r
    sns.barplot(x=r.values, y=r.index, ax=a, palette="viridis", hue=r.index, legend=False)
    a.set_xlabel("Attrition rate (%)"); a.set_title(f"Attrition rate by {c}")
plt.tight_layout(); plt.savefig("fig5_rates.png", dpi=110); plt.show()
for c, r in rates.items():
    print(f"\nAttrition rate (%) by {c}:\n{r.round(1).to_string()}")
```

Output:

```
Attrition rate (%) by OverTime:
OverTime
Yes    30.5
No     10.4

Attrition rate (%) by MaritalStatus:
MaritalStatus
Single      25.5
Married     12.5
Divorced    10.1

Attrition rate (%) by BusinessTravel:
BusinessTravel
Travel_Frequently    24.9
Travel_Rarely        15.0
Non-Travel            8.0

Attrition rate (%) by JobRole:
JobRole
Sales Representative         39.8
Laboratory Technician        23.9
Human Resources              23.1
Sales Executive              17.5
Research Scientist           16.1
Manufacturing Director        6.9
Healthcare Representative     6.9
Manager                       4.9
Research Director             2.5
```

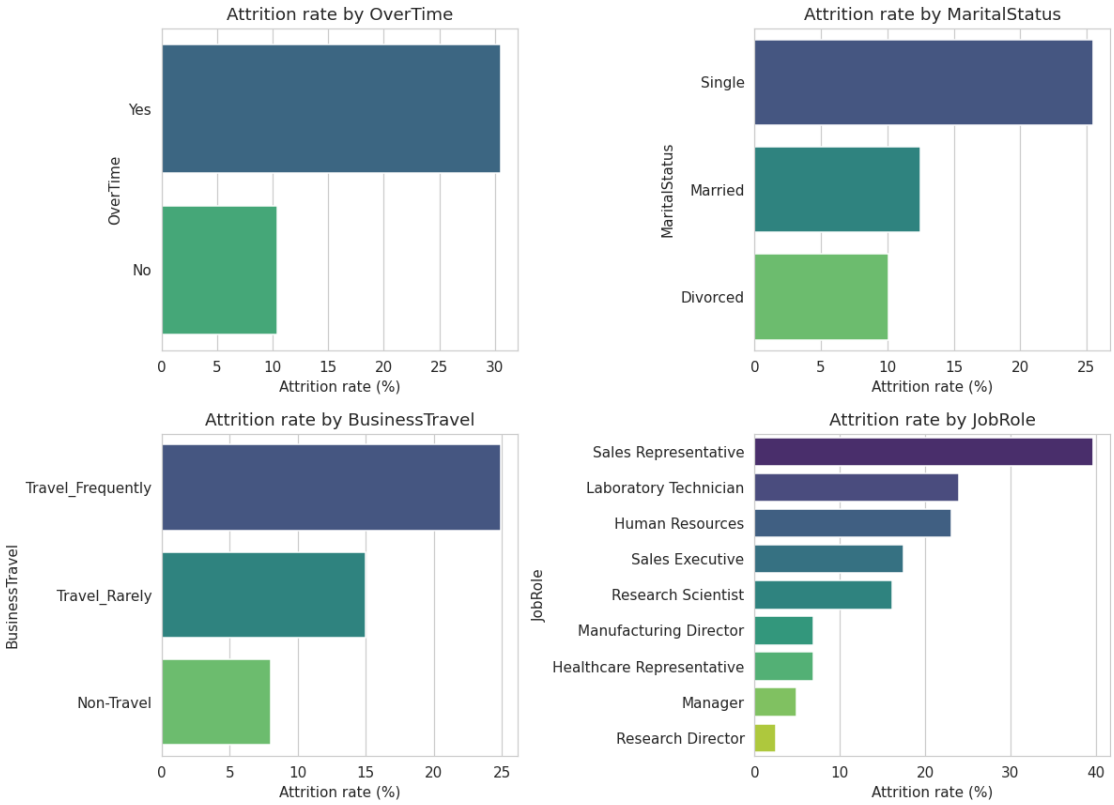


Fig. 5: Attrition rate by category (bivariate)

Step 8: Multivariate analysis

```
# ---------- Step 8b: multivariate analysis ----------
pivot = df.pivot_table(values="Attrition_flag", index="JobLevel", columns="OverTime", aggfunc="mean") * 100
plt.figure(figsize=(5.5, 4))
sns.heatmap(pivot, annot=True, fmt=".1f", cmap="Reds")
plt.title("Attrition rate (%) by JobLevel and OverTime")
plt.tight_layout(); plt.savefig("fig6_multi.png", dpi=110); plt.show()
print("\nAttrition rate (%) by JobLevel x OverTime:\n", pivot.round(1))
print("\nAttrition rate (%) by StockOptionLevel:\n",
      (df.groupby("StockOptionLevel")["Attrition_flag"].mean() * 100).round(1).to_string())
```

Output:

```
Attrition rate (%) by JobLevel x OverTime:
 OverTime    No   Yes
JobLevel
1         15.8  52.6
2          6.7  17.8
3         12.3  20.6
4          2.7   9.1
5          3.9  16.7

Attrition rate (%) by StockOptionLevel:
 StockOptionLevel
0    24.4
1     9.4
2     7.6
3    17.6
```

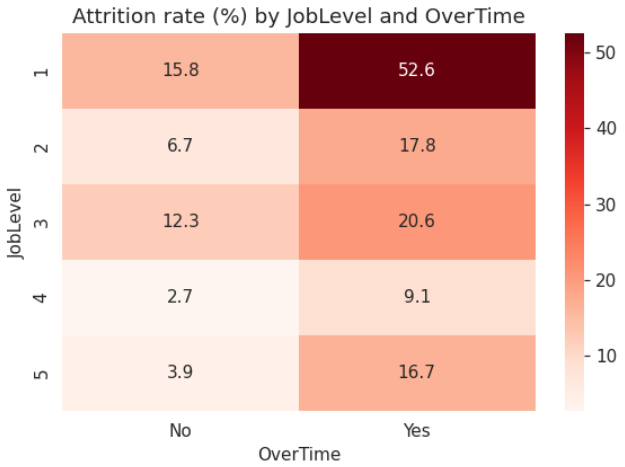


Fig. 6: Attrition rate (%) by JobLevel and OverTime (multivariate)

Key Observations (Step 9)

Area	Finding (from the output above)
Data quality	1,470 rows, 35 columns; no missing values, no duplicates; 3 constant columns (EmployeeCount, Over18, StandardHours) and the ID EmployeeNumber carry no information and were dropped.
Class imbalance	Only 237 of 1,470 employees left (16.1%) vs 1,233 who stayed (83.9%). Accuracy alone would be misleading for a model.
Distributions	MonthlyIncome (skew 1.37, mean 6,503 > median 4,919, 114 IQR outliers) and YearsAtCompany (skew 1.76) are right-skewed; Age is nearly symmetric (skew 0.41).
Workforce mix	Research & Development 961, Sales 446, HR 63. Largest roles: Sales Executive 326, Research Scientist 292, Laboratory Technician 259.
Correlated features	MonthlyIncome-JobLevel r = 0.95; JobLevel-TotalWorkingYears 0.78; MonthlyIncome-TotalWorkingYears 0.77; YearsAtCompany with YearsWithCurrManager 0.77 and YearsInCurrentRole 0.76. These are redundant (multicollinearity).
Link to attrition	Individually weak: TotalWorkingYears -0.17, JobLevel -0.17, MonthlyIncome -0.16, Age -0.16, DistanceFromHome +0.08. Leavers are younger (median age 32 vs 36), earn less (median 3,202 vs 5,204) and have shorter tenure (3 vs 6 years).
OverTime	Attrition 30.5% with overtime vs 10.4% without - the strongest single factor seen.
Other groups	Single 25.5% (Married 12.5%, Divorced 10.1%); frequent travellers 24.9% vs non-travellers 8.0%; Sales Representatives 39.8% vs Research Directors 2.5%; no stock options 24.4% vs 9.4% (level 1) and 7.6% (level 2).
Multivariate	Job level 1 employees working overtime leave at 52.6%, vs 15.8% at level 1 without overtime and 2.7% at level 4 without overtime.

Recommendations (Step 10)

- **Reduce overtime** for junior staff (job level 1): rebalance workload or hire; this group is the highest-risk segment.
- **Review pay and career paths** for Sales Representatives, Laboratory Technicians and HR roles, which show high attrition and lower pay.
- **Limit frequent travel** or compensate it; consider flexible policies for employees who travel often.
- **Focus retention programmes on young, early-tenure, single employees** (mentoring, stock-option eligibility, early promotion).

- **For modelling:** drop constant/ID columns, keep one of each highly correlated group (e.g. JobLevel or MonthlyIncome), handle the 16% class imbalance, and treat the skewed income variable (log transform).

Note: these are associations found by exploratory analysis; they do not by themselves prove causes (see Q6).

Answers to Questions

Q1. What is EDA and why is it performed before machine learning?

Exploratory Data Analysis is the process of examining a dataset using summary statistics and visualisations to understand its structure, distributions, relationships, anomalies and quality before formal modelling (introduced by John Tukey). It is done before machine learning because it:

- reveals missing values, outliers, wrong types and duplicates that would harm a model;
- shows distributions and skewness, guiding transformations and scaling;
- exposes class imbalance (e.g. 16% attrition) and redundant correlated features;
- suggests which variables matter and which new features to create;
- helps select a suitable algorithm and avoids wrong assumptions.

Q2. Differentiate between univariate, bivariate and multivariate analysis.

Type	Variables	Purpose / techniques	Example (HR data)
Univariate	One	Describe distribution: mean, median, histogram, box plot, count plot	Histogram of MonthlyIncome; attrition share 16.1%
Bivariate	Two	Relationship: scatter plot, correlation, grouped bar/box plot, cross-tab	Attrition rate by OverTime (30.5% vs 10.4%); Age vs MonthlyIncome
Multivariate	Three or more	Interactions: heatmap, pair plot, pivot tables, PCA, regression	Attrition by JobLevel x OverTime (52.6% for level 1 with overtime)

Q3. What insights can be obtained from a correlation heatmap?

- Direction and strength of linear relationships between numeric variables (r from -1 to +1; red = positive, blue = negative).
- **Multicollinearity:** highly correlated pairs, e.g. MonthlyIncome and JobLevel (0.95), so one can be dropped.
- Features most related to the target (e.g. TotalWorkingYears -0.17 with Attrition) and variables with almost no relation (JobSatisfaction vs most others near 0).
- Groups of related variables (the tenure variables YearsAtCompany / InCurrentRole / WithCurrManager) and hints for feature selection or dimensionality reduction.
- Limitation: only linear relationships between numeric variables are captured.

Q4. Purpose of histograms, box plots and scatter plots in EDA.

Plot	Purpose	What it shows
Histogram	Show the distribution of one numeric variable	Shape (normal/skewed), spread, modality, central tendency; e.g. right-skewed MonthlyIncome
Box plot	Summarise spread and detect outliers; compare groups	Median, quartiles, IQR, whiskers and outlier points; e.g. income by Attrition
Scatter plot	Show relation between two numeric variables	Trend, linear/non-linear pattern, clusters, outliers; e.g. TotalWorkingYears vs MonthlyIncome

Q5. How can EDA help identify data quality issues?

- `isnull().sum()` and `duplicated().sum()` reveal missing values and duplicates.
- `info()` shows wrong data types (e.g. numbers stored as text).
- `describe()` and box plots reveal impossible or extreme values (negative age, huge income).
- `value_counts()` exposes inconsistent labels ("male", "Male ") and rare categories.
- `nunique()` finds constant columns (EmployeeCount, Over18, StandardHours) and ID-like columns.

- Count plots show class imbalance; histograms show skewness needing transformation.

Q6. Why is correlation important in predictive analytics? Can correlation imply causation?

Correlation measures how strongly two variables move together. It is important because strongly correlated predictors with the target are useful features, while strongly correlated predictors with each other cause redundancy and unstable coefficients. However, **correlation does not imply causation**: the relationship may be due to chance, a hidden (confounding) variable, or reverse direction. For example, ice-cream sales and drowning both rise in summer because of temperature, not because one causes the other. In the HR data, older employees leave less (r = -0.16), but age may not be the cause; experience, level and pay also change with age. Causation needs controlled experiments or causal inference methods.

Q7. Which visualisation would you use for categorical and numerical variables? Justify.

Situation	Best visualisation	Justification
One categorical	Count plot / bar chart (pie only for few categories)	Shows frequency of each category and the most common ones
One numerical	Histogram / KDE plot	Shows distribution, skewness and spread
Categorical vs numerical	Box plot or violin plot (also bar of mean)	Compares median, spread and outliers per group, e.g. MonthlyIncome by Attrition
Categorical vs categorical	Grouped/stacked bar chart, heatmap of rates	Shows rate per group, e.g. attrition rate by OverTime or JobLevel x OverTime
Numerical vs numerical	Scatter plot (colour by category)	Shows the relationship and the effect of a third variable

Q8. What business insights can be derived from the HR dataset through EDA?

- Overall attrition is 16.1%, but it is concentrated in specific groups.
- **Overtime** is the biggest driver: 30.5% attrition vs 10.4%; for job level 1 with overtime it is 52.6%.
- **Role/pay:** Sales Representatives (39.8%) and Laboratory Technicians (23.9%) leave most; leavers have much lower median income (3,202 vs 5,204).
- **Profile:** younger, single, early-tenure employees (median tenure 3 years) and frequent business travellers (24.9%) are at higher risk.
- **Benefits:** employees with no stock options leave at 24.4% vs 7-9% with levels 1-2.
- Actions: cut overtime, review compensation and career paths in high-risk roles, offer stock options and mentoring, and monitor risk segments.

(For Netflix: content type mix, release-year growth, country and genre trends; for Superstore: sales/profit by region, category and discount.)

Q9. How does EDA contribute to feature selection and model building?

- Identifies **relevant features** (strong relationship with the target, e.g. OverTime, JobLevel) and **irrelevant ones** (constant, ID columns).
- Detects **redundant features** through correlation (keep one of MonthlyIncome / JobLevel).
- Guides **preprocessing**: imputation, outlier handling, log transform of skewed features, scaling, encoding choice.
- Inspires **feature engineering** (income bands, tenure ratios, overtime x level interaction).
- Reveals **class imbalance** so that stratified splitting, class weights/SMOTE and F1/recall metrics are chosen.
- Helps choose model type (linear vs tree-based) depending on linearity and interactions.

Q10. Challenges while performing EDA on large-scale real-world datasets?

- **Volume:** data may not fit in memory; plotting millions of points is slow and cluttered (use sampling, aggregation, Dask/Spark).
- **High dimensionality:** hundreds of features make heatmaps and pair plots unreadable (use feature selection, PCA).
- **Data quality:** many missing values, inconsistent formats, duplicates and noise across sources.
- **Mixed and unstructured data types:** text, dates, images and categorical variables with many levels.
- **Sampling bias and imbalance** can mislead conclusions; spurious correlations appear with many variables.
- **Time and computation cost**, privacy/security constraints and interpreting results correctly (correlation vs causation).

Conclusion

EDA on the IBM HR Employee Attrition dataset showed a clean but imbalanced dataset (16.1% attrition) with skewed income and tenure variables and several highly correlated features. Attrition is strongly linked to overtime, low job level, low pay, frequent travel, single marital status and lack of stock options. These insights support targeted retention actions and guide preprocessing, feature selection and model building.